PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Emmanuel John
Program/Batch	B084-Networkwalks
Date	09 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.
All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain’s name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.1.4** and **WP Download Manager 3.3.76**, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

I used **Curl** with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint /wp-json/.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified four live hosts:

- 172.20.10.1
- 172.20.10.8

The example results also included one MAC addresses.

- 36:FE:77:78:DB:64

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

Note: The actual subnet, number of hosts and addresses should be replaced with the results from my own network when submitting the report.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● **Medium**
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

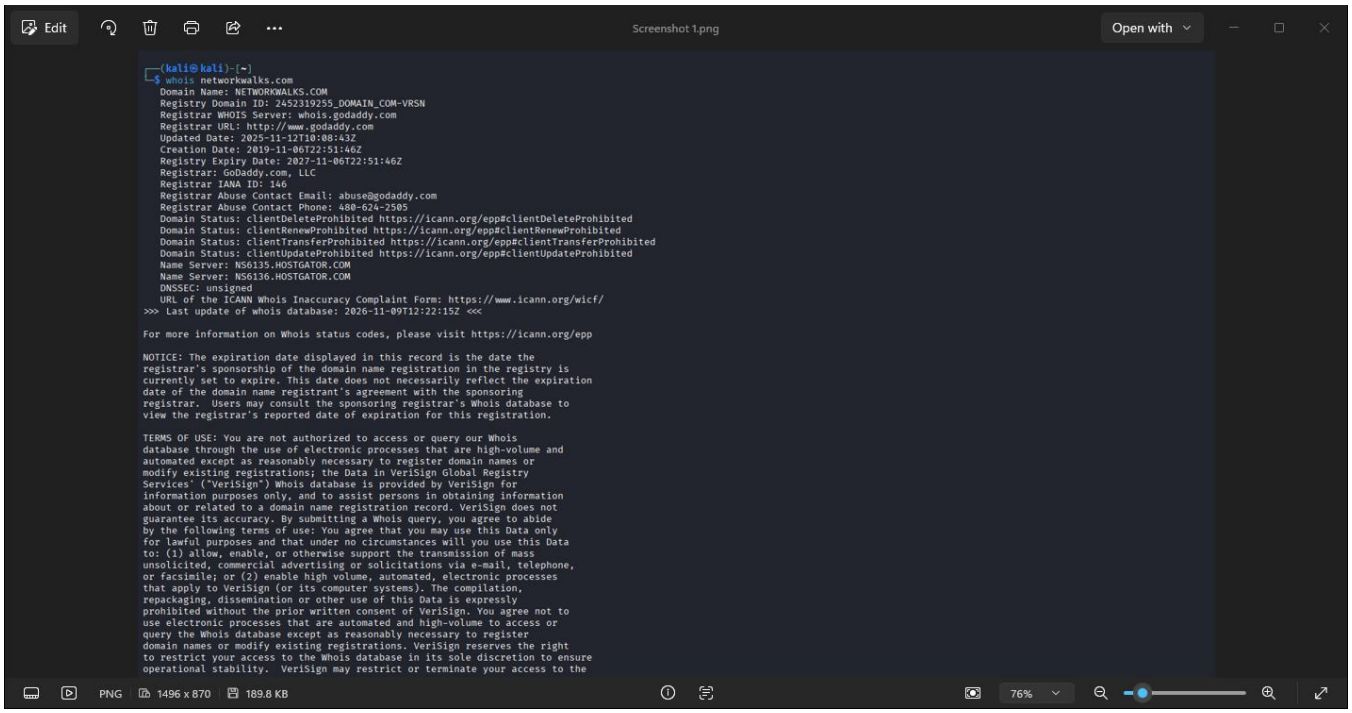
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

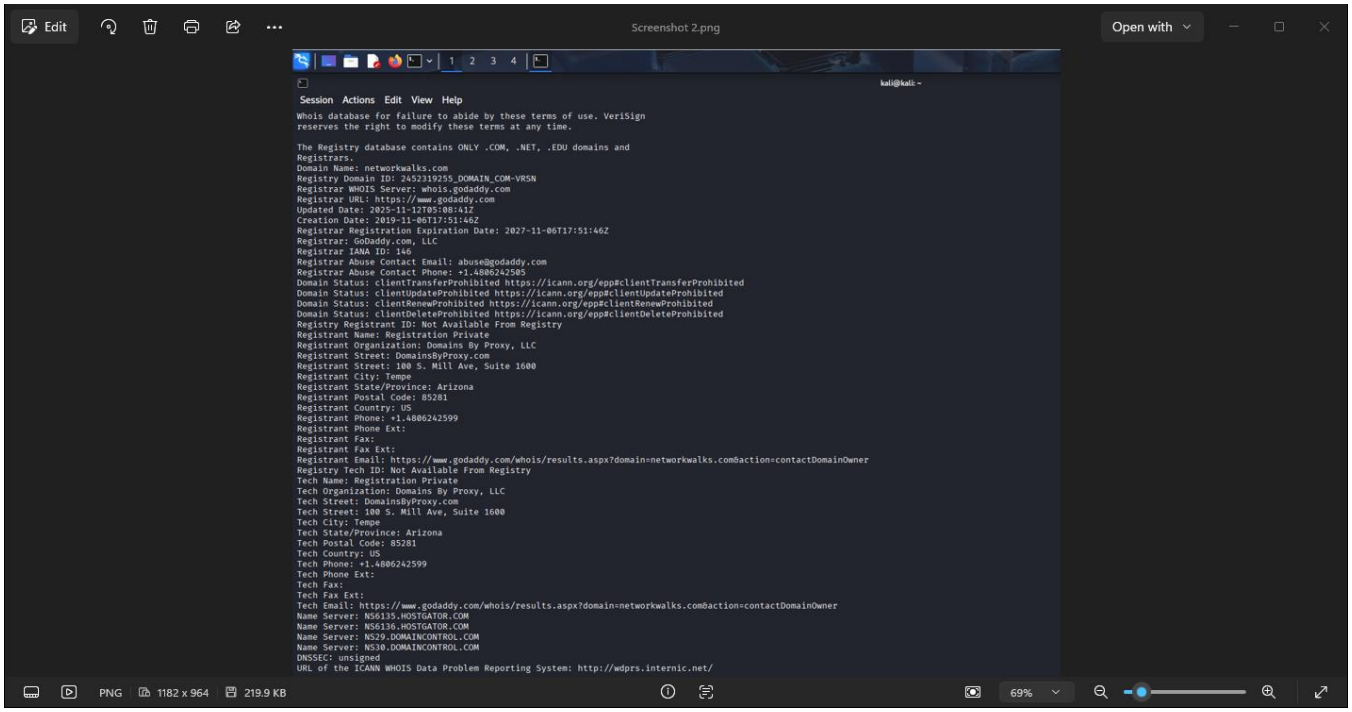
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

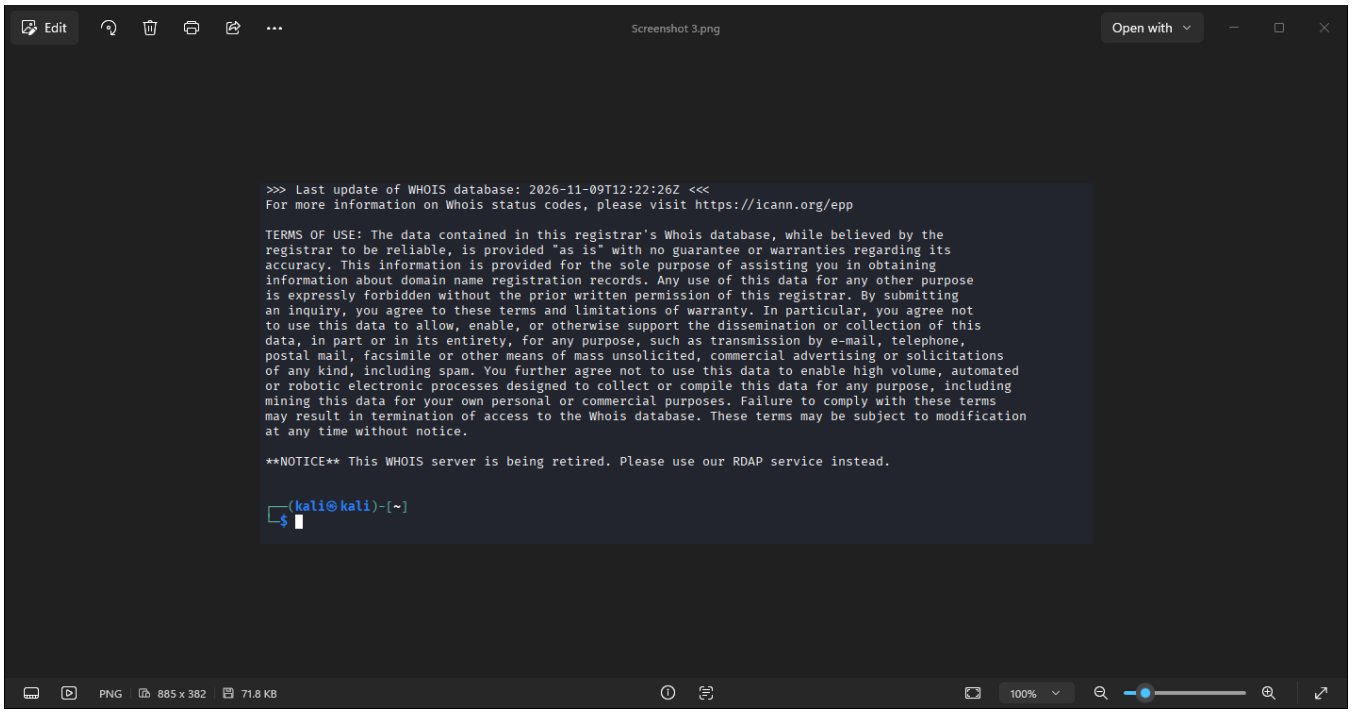
Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

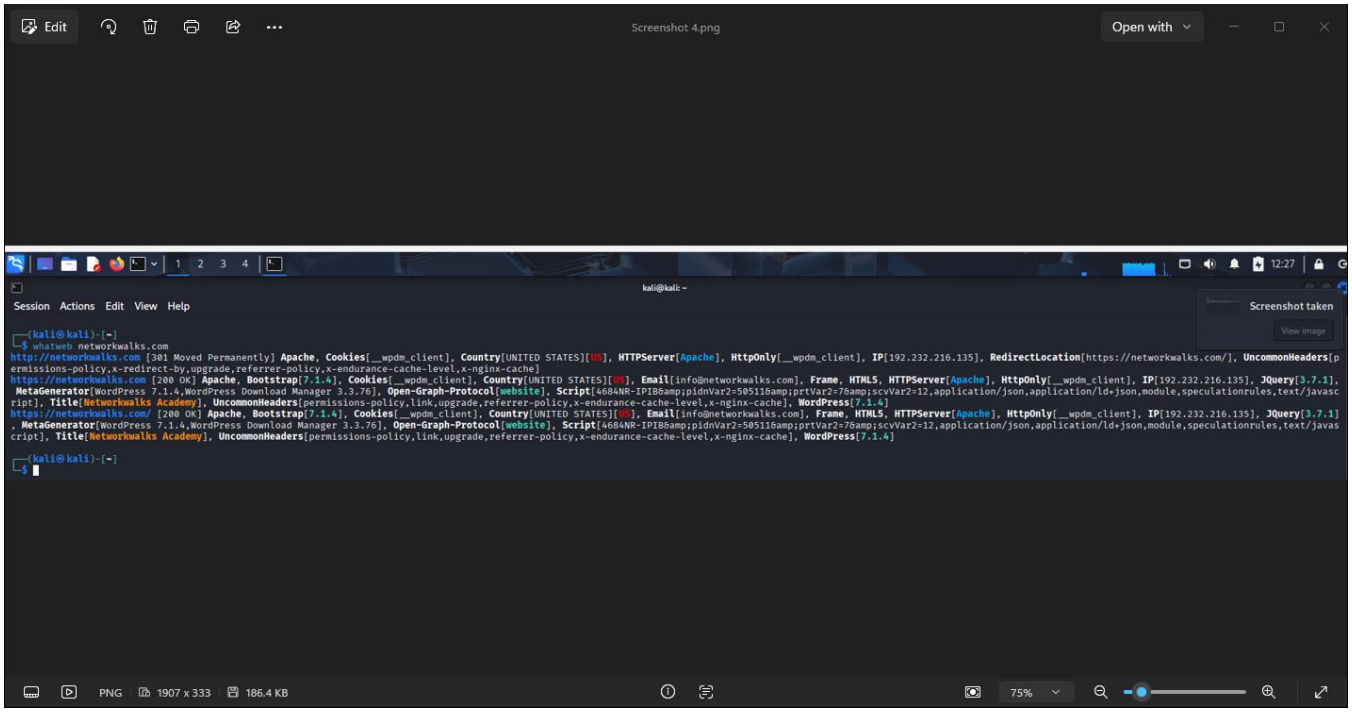
Whois



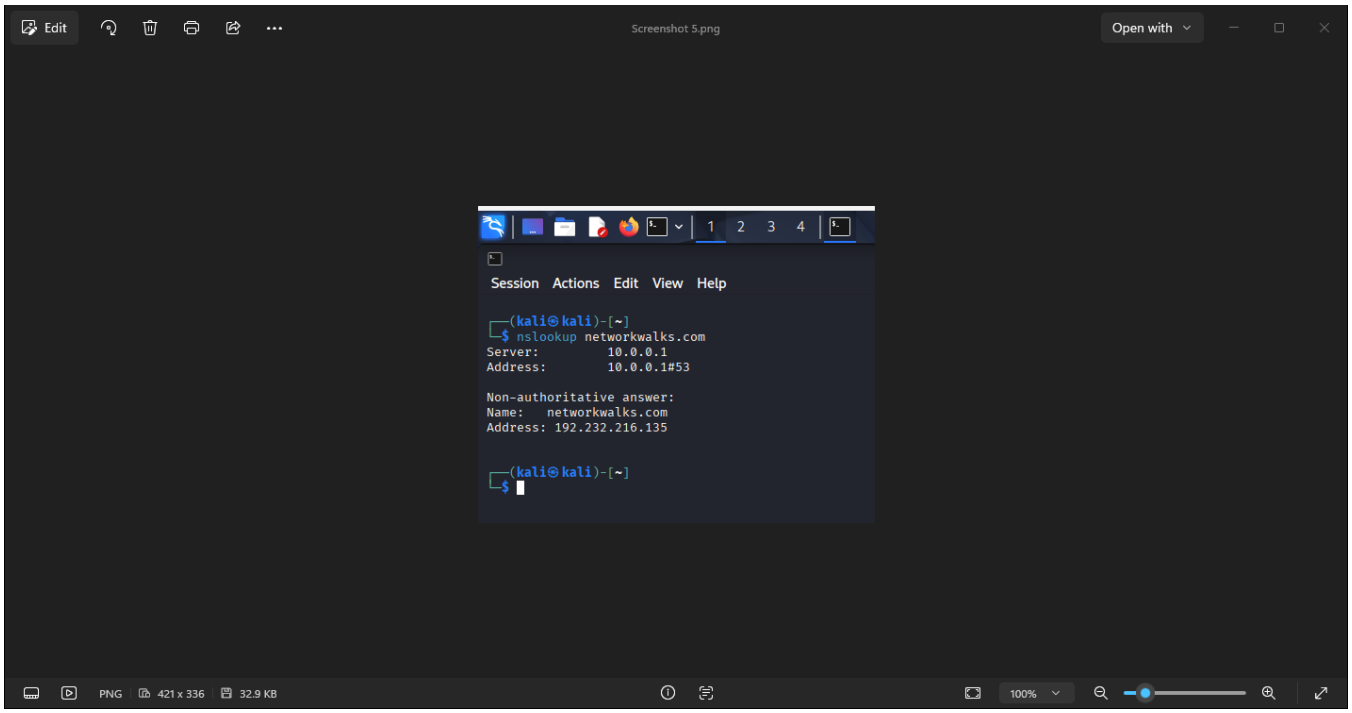



Edit
Screenshot 3.png
Open with
>>> Last update of WHOIS database: 2026-11-09T12:22:26Z <<<
For more information on Whois status codes, please visit https://icann.org/epp
TERMS OF USE: The data contained in this registrar's Whois database, while believed by the registrar to be reliable, is provided "as is" with no guarantee or warranties regarding its accuracy. This information is provided for the sole purpose of assisting you in obtaining information about domain name registration records. Any use of this data for any other purpose is expressly forbidden without the prior written permission of this registrar. By submitting an inquiry, you agree to these terms and limitations of warranty. In particular, you agree not to use this data to allow, enable, or otherwise support the dissemination or collection of this data, in part or in its entirety, for any purpose, such as transmission by e-mail, telephone, postal mail, facsimile or other means of mass unsolicited, commercial advertising or solicitations of any kind, including spam. You further agree not to use this data to enable high volume, automated or robotic electronic processes designed to collect or compile this data for any purpose, including mining this data for your own personal or commercial purposes. Failure to comply with these terms may result in termination of access to the Whois database. These terms may be subject to modification at any time without notice.
NOTICE This WHOIS server is being retired. Please use our RDAP service instead.
(kali㉿kali)-[~]
$
PNG
885 x 382
71.8 KB
100%

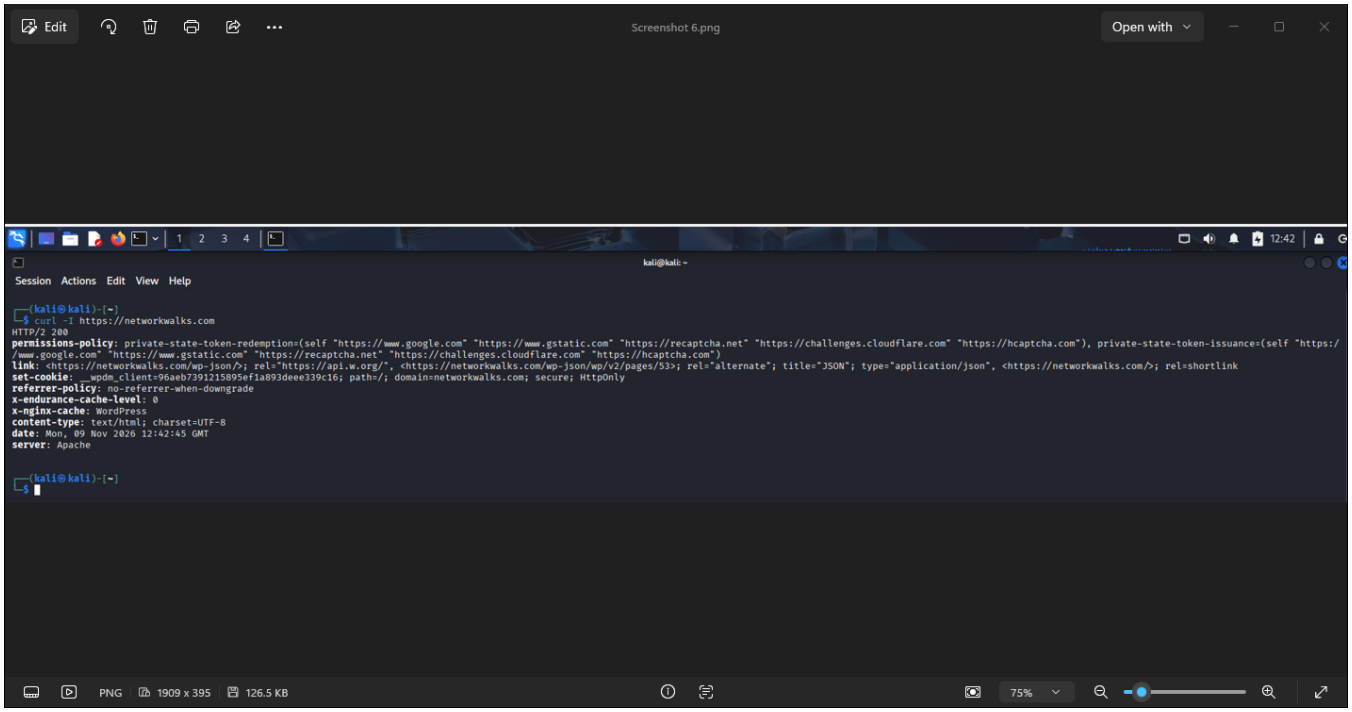
WhatWeb:


Edit
Screenshot 4.png
Open with
kali@kali: ~
Session Actions Edit View Help
Screenshot taken
View image
(kali㉿kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB6amp;pidnVar2=505116amp;prtVar2=76amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB6amp;pidnVar2=505116amp;prtVar2=76amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
(kali㉿kali)-[~]
$
PNG
1907 x 333
186.4 KB
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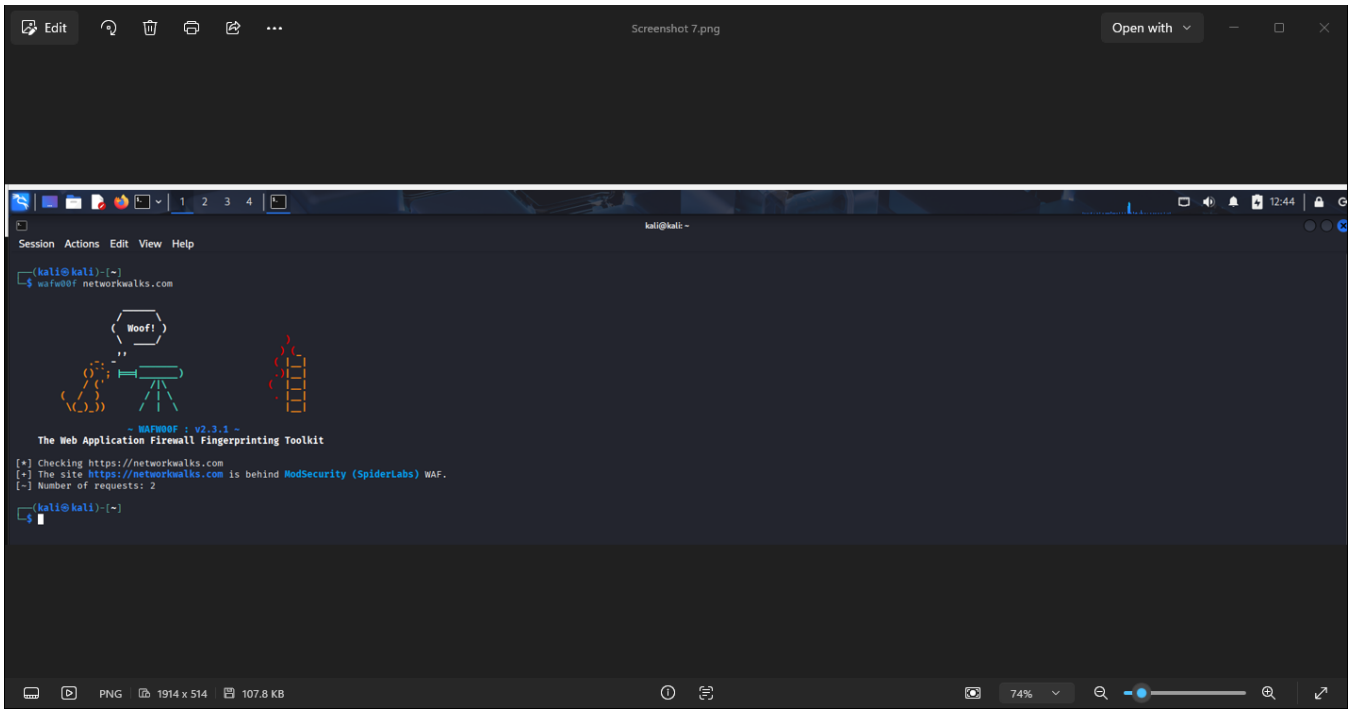
NsLookup:



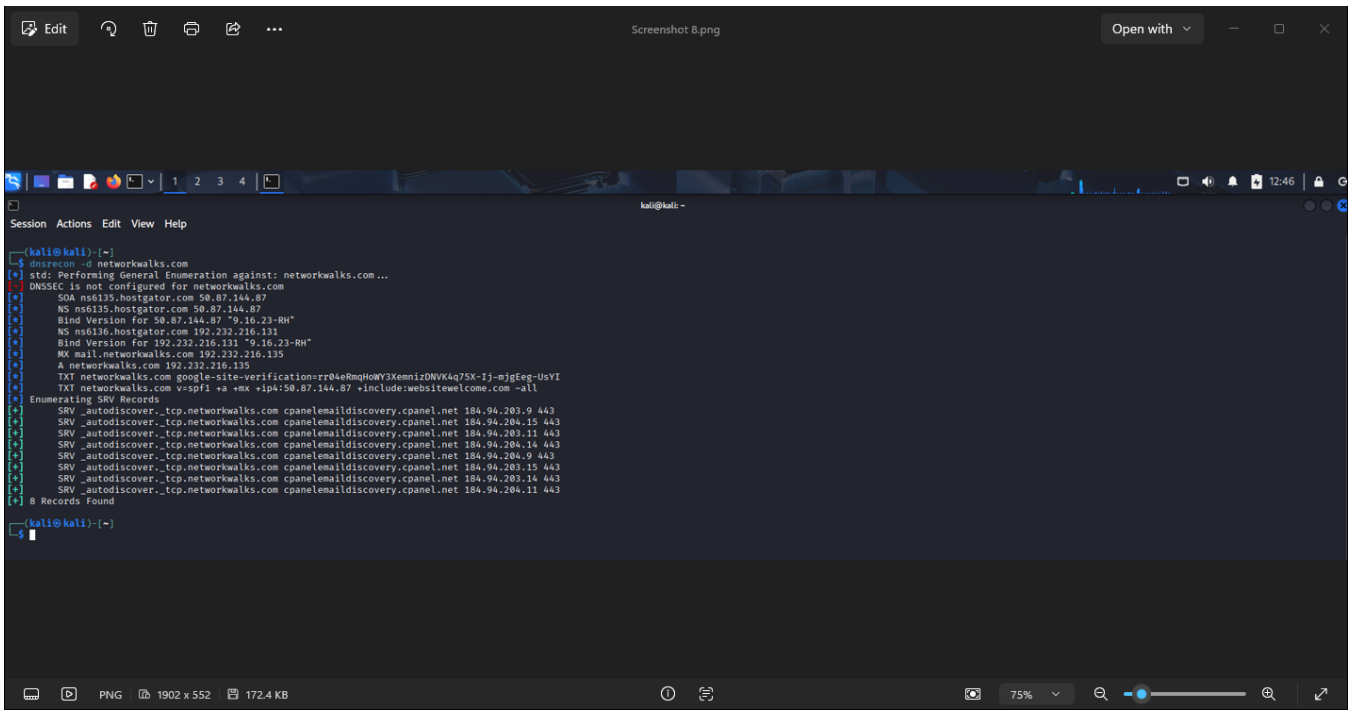
Curl:



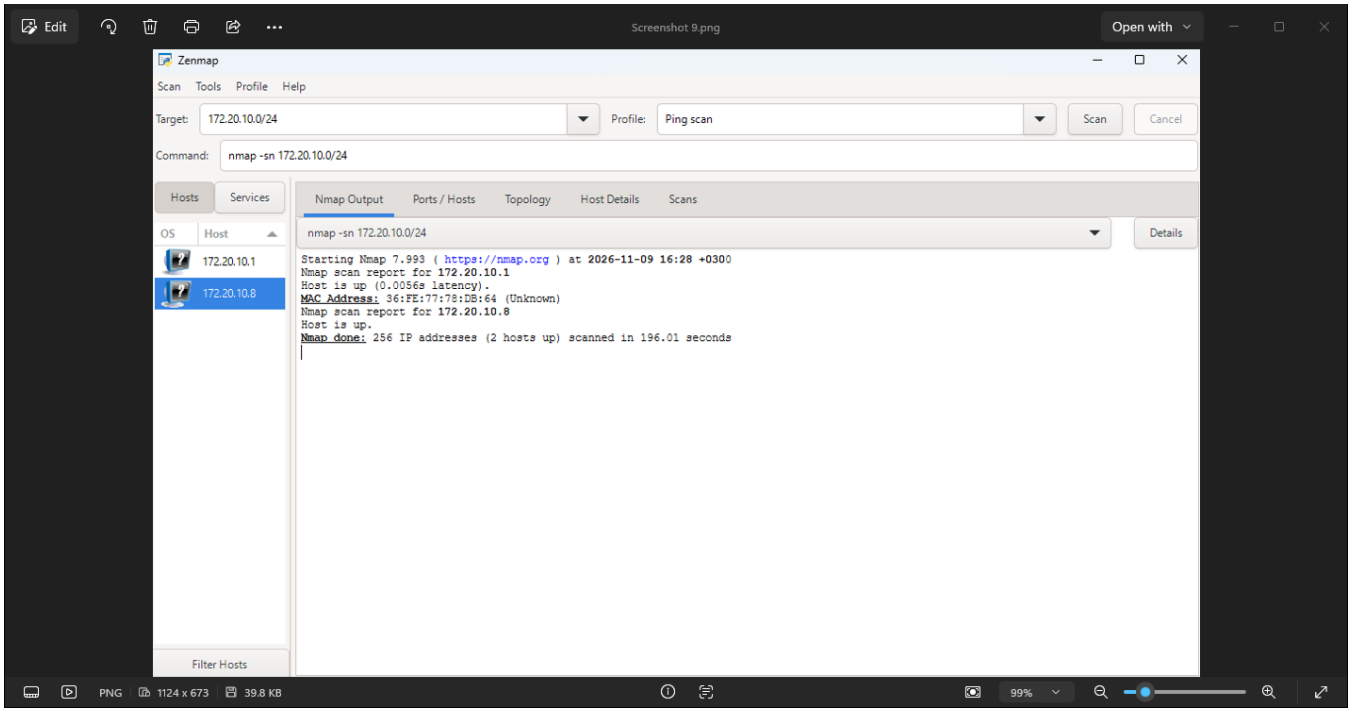
Wafw00f:

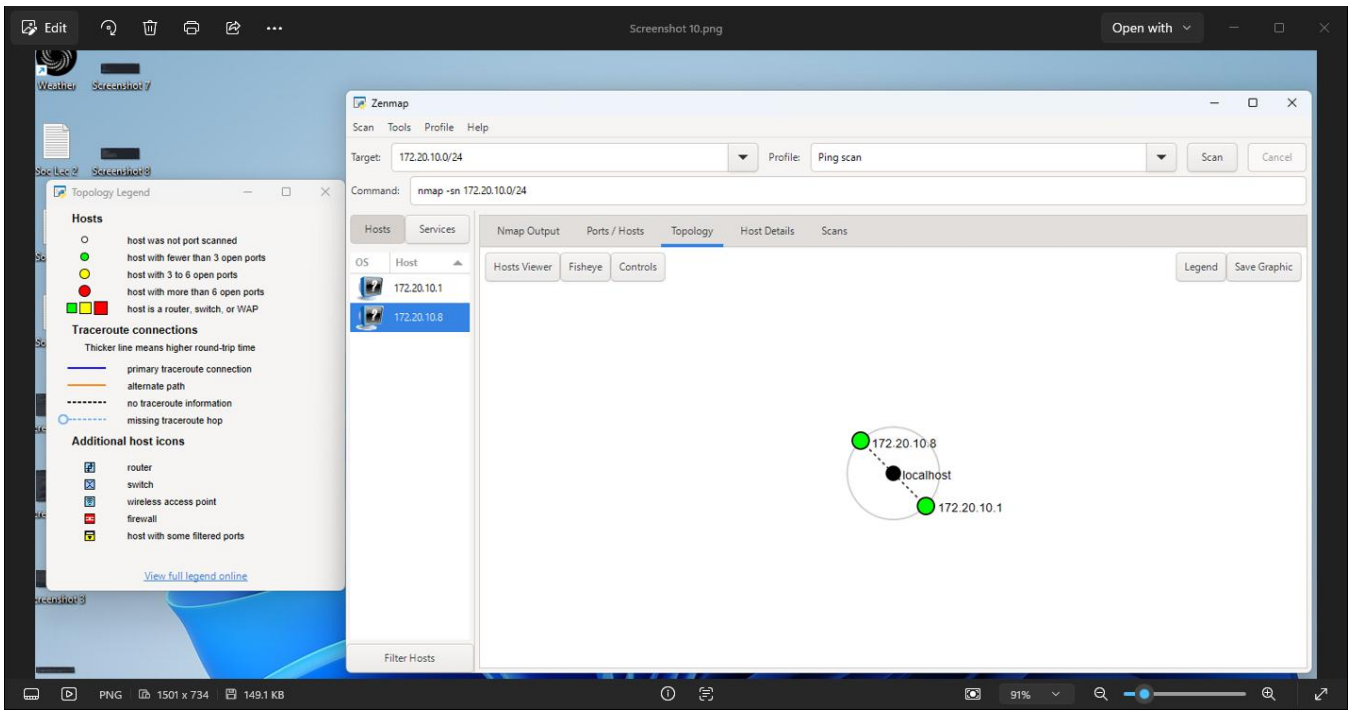


DNSrecon:



Zenmap:


Edit
Screenshot 9.png
Open with
Zenmap
Scan Tools Profile Help
Target: 172.20.10.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 172.20.10.0/24
Hosts
Services
OS
Host
172.20.10.1
172.20.10.8
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 172.20.10.0/24
Details
Starting Nmap 7.993 (https://nmap.org) at 2026-11-09 16:28 +0300
Nmap scan report for 172.20.10.1
Host is up (0.0056s latency).
MAC Address: 36:FE:77:78:DB:64 (Unknown)
Nmap scan report for 172.20.10.8
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 196.01 seconds
Filter Hosts
PNG
1124 x 673
39.8 KB
99%


Edit
Screenshot 10.png
Open with
Topology Legend
Hosts
host was not port scanned
host with fewer than 3 open ports
host with 3 to 6 open ports
host with more than 6 open ports
host is a router, switch, or WAP
Traceroute connections
Thicker line means higher round-trip time
primary traceroute connection
alternate path
no traceroute information
missing traceroute hop
Additional host icons
router
switch
wireless access point
firewall
host with some filtered ports
View full legend online
Zenmap
Scan Tools Profile Help
Target: 172.20.10.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 172.20.10.0/24
Hosts
Services
OS
Host
172.20.10.1
172.20.10.8
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Hosts Viewer
Fisheye
Controls
Legend
Save Graphic
172.20.10.8
localhost
172.20.10.1
Filter Hosts
PNG
1501 x 734
149.1 KB
91%

-End-

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📌 **Project Information**
Program Name: Footprint and Network Scanning | **Week:** 02 | **Repository:** GitHub